

Day 17 (26/11/2025)

Data Structures -

Data structures is a specialized format for organizing, storing and managing data in a computer's memory or storage.

The purpose of data structures is to enable efficient access, modification, & manipulation of data, making programs more effective & performant.

Different types of Data structures -

Python offers both built-in Data structures and the capability to create user-defined data structures.

1) Built-in Data structures - These are fundamental data types provided directly by python.

list, tuple, set, dict, str.

List - it is multivalued heterogenous (we can pass different data types) data collectors,

- it's ordered, mutable (adding, updating, deleting) and it will allow duplicates

- it's represented by `[]` or `list()`

```
l1 = [25, "poojyy", 38.9, True, 490]
```

l1
o/p => [25, 'poojyy', 38.9, True, 490]

1. $l_1 = [25, "poojyy", 38.9, True, 490]$

$l_1[0]$

o/p $\Rightarrow 25$

Operations in list -

1) $append()$ $\rightarrow$ Adds a new element at the end of the list.

Ex- $l_1 = [25, "poojyy", 38.9, True, 490]$

$l_1.append(25)$ # add

l_1

o/p $\Rightarrow [25, 'poojyy', 38.9, True, 490, 25]$

2) $update()$ _{assignment}

we update list values by assigning a new value to an index.

~~$l_1 = f$~~

Ex- $l_1[2] = 100$

o/p $\Rightarrow [25, 'poojyy', 100, True, 490, 25]$

3) $Insert()$

Adds element at a specific index.

Ex- $l_1.insert(4, False)$ # without deleting just inserting the value.

l_1

o/p $\Rightarrow [25, 'poojyy', 100, True, False, 490, 25]$

4) pop() -

removes an element by index and returns it.
By default, pop() removes last element.

l1 = [25, 'poojyy', 100, True, False, 490, 25]

l1.pop(4)

l1

o/p → [25, 'poojyy', 100, True, 490, 25]

5) remove() -

remove the first occurrence of a specific value.

l1.remove(25)

l1

o/p → [25, 'poojyy', 100, 490, 25]

for i in l1:

print(i)

25
poojyy
100
490
25

Q) l2 = [1, 2, 5, 7, 8]

l3 = [10, 20, 30, 60, 30]

o/p - [11, 22, 15, 67, 38]

s1 = []

for i in range(0, 5, 1):

s = l2[i] + l3[i]

s1.append(s)

o/p → [11, 22, 15, 67, 38]

print(s1)

a) create even, odd, prime numbers list from 1 to 20 numbers.

```
el = []
```

```
ol = []
```

```
pl = []
```

```
for i in range(1, 21, 1):
```

```
    if (i % 2 == 0):
```

```
        el.append(i)
```

```
    else:
```

```
        ol.append(i)
```

```
    if (i != 1):
```

```
        for k in range(2, i, 1):
```

```
            if i % k == 0:
```

```
                break
```

```
        else:
```

```
            pl.append(i)
```

```
print(el)
```

```
print(ol)
```

```
print(pl)
```

o/p ->

[2, 4, 6, 8, 10, 12, 14, 16, 18, 20]

[1, 3, 5, 7, 9, 11, 13, 15, 17, 19]

[2, 3, 5, 7, 11, 13, 17, 19].

Q) $a = [1, 2, 3, 4, 5]$

$b = [4, 5, 6, 7, 8]$

o/p $\Rightarrow$ ① $[1, 2, 3, 4, 5, 6, 7, 8]$

② $[1, 2, 3, 6, 7, 8]$

③ $[4, 5]$

④ $[1, 2, 3]$

⑤ $[6, 7, 8]$

1) o/p

$a = [1, 2, 3, 4, 5]$

$b = [4, 5, 6, 7, 8]$

$c = a$

for i in b :

if i not in a :

$c.append(i)$

print(c)

$[1, 2, 3, 4, 5, 6, 7, 8]$

o/p

2) o/p

$a = [1, 2, 3, 4, 5]$

$b = [4, 5, 6, 7, 8]$

$c = []$

for i in a :

if i not in b :

$c.append(i)$

for j in b :

if j not in a :

$c.append(j)$

print(c)

o/p $[1, 2, 3, 6, 7, 8]$

3) o/p

$a = [1, 2, 3, 4, 5]$

$b = [4, 5, 6, 7, 8]$

$c = []$

for i in a :

if i in b :

$c.append(i)$

print(c)

o/p $[4, 5]$

4) o/p

$a = [1, 2, 3, 4, 5]$

$b = [4, 5, 6, 7, 8]$

$c = []$

for i in a :

if i not in b :

$c.append(i)$

print(c).

o/p $[1, 2, 3]$

c) o/p

$a = [1, 2, 3, 4, 5]$

$b = [4, 5, 6, 7, 8]$

$c = []$

for i in b :

if i not in a :

print(c) $c.append(i)$

o/p
 $[6, 7, 8]$